

AI Emergency Patient Intake

Team Workflow and Role Ownership

Implementation merged; integration and demo-readiness checkpoint | v5 | 8 Sep 2026

Product goal

Reduce the information bottleneck between a distressed patient and emergency medical staff by converting natural conversation, caregiver input, staff observations and verified history into structured clinician-ready information.

ROLE 1 Voice AI Engineer	ROLE 2 AI / Agent Engineer	ROLE 3 Backend Engineer	ROLE 4 Database & Integration	ROLE 5 Frontend Engineer
Live audio, transcription, spoken replies, turn-taking and interruption handling.	prompts, conversation state, information extraction, agent decisions and safety guardrails.	FastAPI, tool execution, authentication, workflow orchestration and external actions.	Patient records, medical history, call logs, permissions, privacy and data retrieval.	Live dashboard, structured summary, alerts, staff review and final approval interface.
SSE publisher		SSE receiver		
SOHA RAEES	FAZWAN ZAINUDDIN	MOZZAM SHAHID	ISHAAN SAMA	JONATHAN
ONE OWNER - CLEAR HANDOFFS	ONE OWNER - CLEAR HANDOFFS	ONE OWNER - CLEAR HANDOFFS	ONE OWNER - CLEAR HANDOFFS	ONE OWNER - CLEAR HANDOFFS

8 SEP UPDATE - ALL FIVE ROLE COMPONENTS ARE NOW ON MAIN

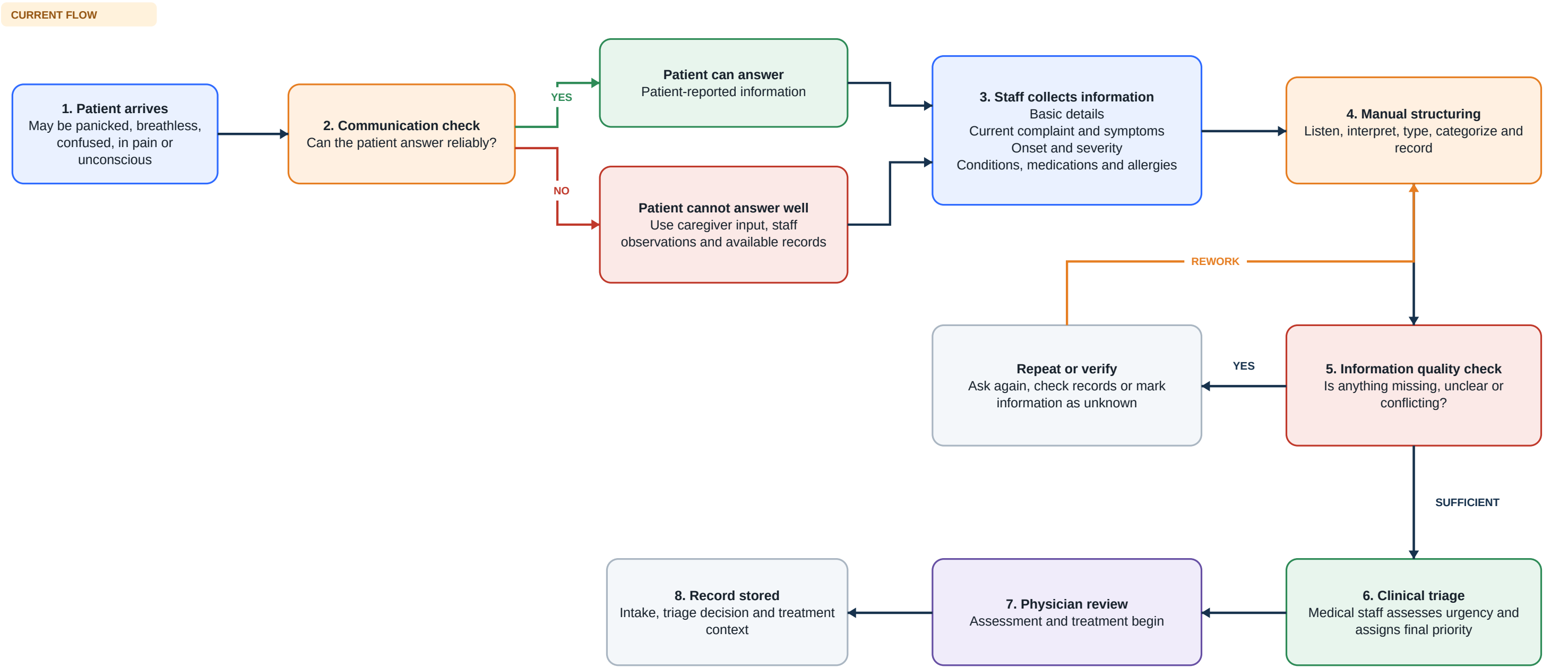
Verified locally: backend, voice demo service and dashboard run together; a synthetic case was extracted and accepted by Role 3.

Next: rotate exposed keys, configure Supabase securely, close domain review and complete the recorded end-to-end demo and rehearsal.

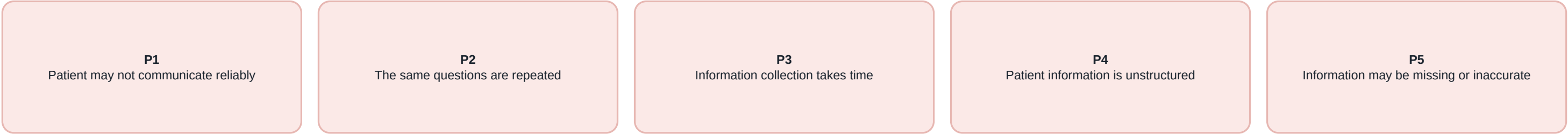
All five role components are on main. See pages 5-6 for verified evidence, secure setup and the remaining demo gates.

1. Current Emergency Intake Flow

Where the communication and documentation bottlenecks occur



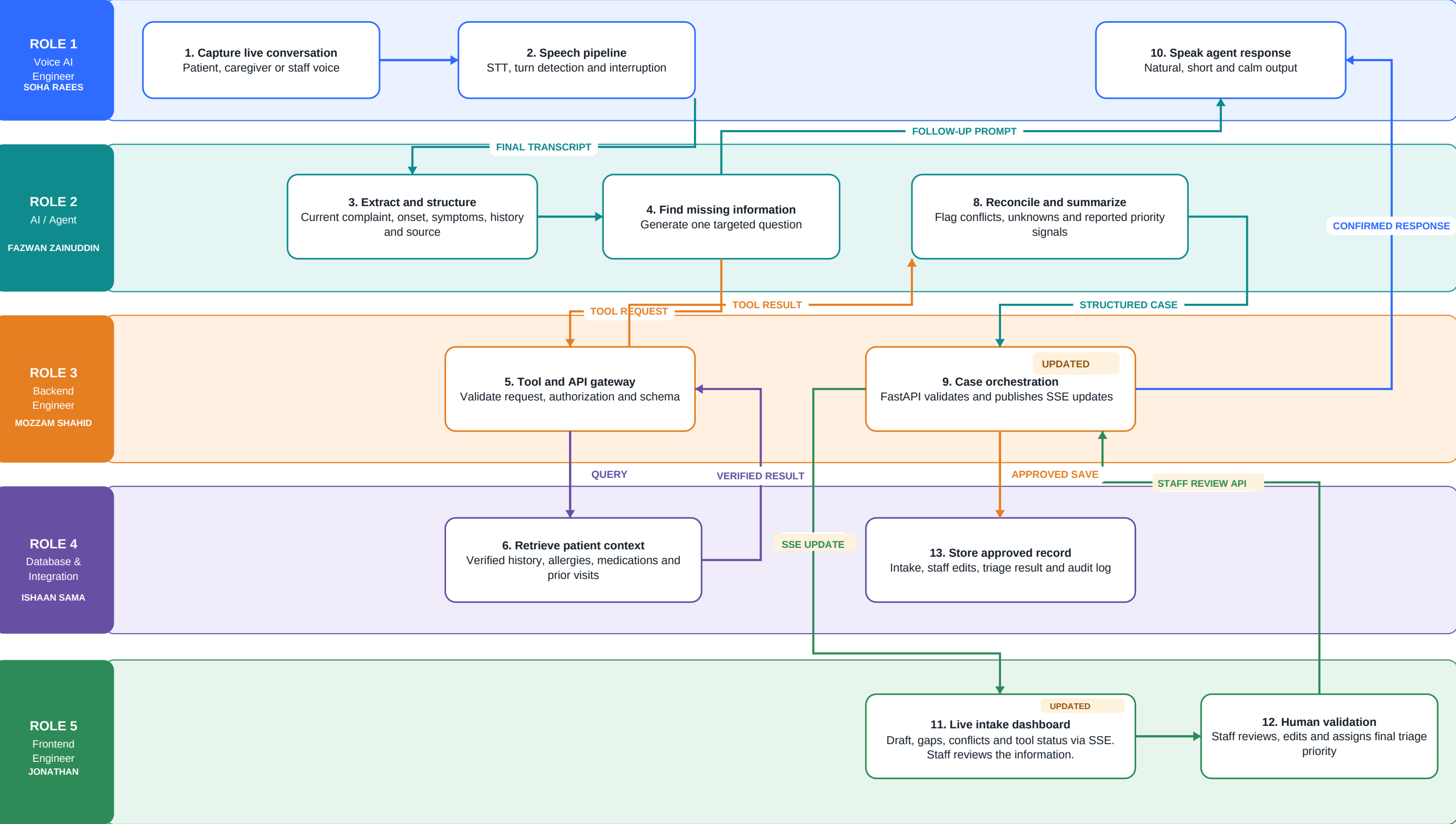
Pain points identified



2. Proposed AI-Assisted Workflow by Role

Each lane has one owner; arrows define the integration contracts

CURRENT HANDOFF: step 9 publishes SSE updates to step 11; staff review returns through an API. Lane order and step numbers are unchanged.



Human decision gate: AI structures and highlights reported information. Emergency medical staff remain responsible for clinical assessment and final triage priority.

3. Ownership and Handoff Contract

Use this page to prevent duplicate work and integration conflicts

Working rule: Every role owns its internal implementation, but all cross-role data must follow an agreed contract: event name, JSON schema, success response, error response and owner.

UPDATED CELLS Highlighted notes clarify the current SSE handoff. Role 1 is assigned to Soha Raees.

Role	Owns and delivers	Sends to other roles	Receives from other roles	Must not duplicate
Role 1 Voice AI Engineer SOHA RAEES	AssemblyAI connection; microphone/audio streaming; STT/TTS; turn detection; interruptions; session state.	Final transcript turns, session ID/status and spoken-response completion.	Agent response text, conversation state and approved prompt configuration from Role 2.	Triage logic, backend business logic, database schema or dashboard UI.
Role 2 AI / Agent Engineer FAZWAN ZAINUDDIN	System prompt; conversation flow; extraction schema; missing-information logic; source/confidence labels; tool policy; safety guardrails; structured summary.	CURRENT Response text to Role 1; tool requests and structured extraction to Role 3. The backend owns the SSE event wrapper.	Final transcript from Role 1; verified tool result from Role 3; retrieved context through backend tools.	Direct database access, API implementation, UI code or the medical staff's final triage decision.
Role 3 Backend Engineer MOZZAM SHAHID	CURRENT FastAPI; authentication; validation; tool execution; workflow orchestration; SSE publishing and reconnection; staff-review APIs.	CURRENT Queries and approved saves to Role 4; live SSE updates to Role 5; actual tool results to Role 2.	Validated tool requests from Role 2; data responses from Role 4; staff actions from Role 5.	Agent prompt/decision policy, physical data model ownership or frontend rendering.
Role 4 Database & Integration ISHAAN SAMA	Patient schema; synthetic records; history and allergies; call logs; audit trail; permissions; privacy; retention; query performance.	Verified records, not-found responses, conflicts and save confirmations to Role 3.	Authorized read/write requests from Role 3 and approved staff updates routed by backend.	Calling the LLM directly, frontend UI, voice pipeline or backend workflow decisions.
Role 5 Frontend Engineer JONATHAN	Live monitoring dashboard; structured intake; source/confidence display; missing/conflict flags; staff edit, review and approval interface.	Staff corrections, approval, final triage selection and action requests to Role 3.	CURRENT SSE updates with the draft, missing information, conflicts and tool status from Role 3; confirmed review/save responses.	Direct database writes, independent triage logic, voice pipeline or backend tool execution.

Source: Emergency Patient Intake - Problem Analysis & AI Opportunity Assessment. This is an architecture proposal for team agreement, not a clinical protocol.

4. Implementation Status and Demo Readiness

CURRENT v5

8 September update - all five role components are on main; integration, security and presentation gates remain.

Current implementation: AssemblyAI-compatible voice + Role 2 extraction + FastAPI/SSE + Supabase/SQLite + Next.js dashboard.

Readiness rule: component delivery is complete; competition readiness requires secure keys, domain sign-off and a recorded end-to-end rehearsal.

Verified implementation state

Area	Evidence on 8 September	Meaning
Roles 1-5	All implementation paths are on main; Role 1-5 delivery issues are closed.	Component delivery is complete; PR #34 contains the final integration improvements.
Local verification	49 Python tests, 6 dashboard tests, contract validation and Next.js production build pass.	Backend :8000, voice demo :8001 and dashboard :3000 can run together.
Hosted services	Team evidence reports AssemblyAI and Supabase testing; Fazwan's machine has no local .env yet.	Hosted access is not ready on this machine until fresh keys are configured securely.

Readiness actions in order

Priority / owner	Action	Completion evidence
1 - Security / key owners	Do not merge PR #35 with .env. Rotate exposed keys; address the two high npm audit findings before public deployment.	Old keys revoked; clean PR and secret scan pass; dependency upgrade or mitigation is documented.
2 - Ishaan + Fazwan	Create a new local .env from .env.example; use the project URL and fresh server-side key. Never commit it.	python -m src.data --check-supabase reports OK without printing credentials.
3 - Role 4 + Role 3	Confirm the Role 4 migration and tables in Supabase Table Editor; run one synthetic lookup and approved save.	Patient lookup, case, review, approved record and audit row are visible exactly once.
4 - All technical owners	Run backend, voice and dashboard together; complete one synthetic intake-to-review-to-save flow.	Issue #7 has timestamps, case ID, screenshots/log summary and known limitations.
5 - Domain + Mariam	Close clinical/safety decisions, prepare fallback screenshots/recording and run a timed rehearsal.	Issues #6 and #11 are closed with sign-off, script, timing and action items.

NEXT TEAM CHECKPOINT - DEMO READINESS, NOT MORE FEATURE BUILDING

Use the merged implementation as-is. Focus on secure configuration, one repeatable synthetic scenario, human-review evidence and a backup demo path.

Do not use real patient data. AI output remains an unverified draft; authorised staff retain the final clinical decision. Domain approval rules must be recorded before the competition demo.

Stable run guide: GETTING_STARTED.md. Supabase is optional for local development; SQLite is the safe fallback. Hosted persistence requires the migration and fresh keys.

Technical reference: [FastAPI SSE documentation](#). Verified against main commit aa95e0c on 8 September 2026.

5. Confirmed Roles, Delivered Components and Supabase

All role implementations are now on main. Ownership remains unchanged; the team now moves to integration and demo readiness.

CURRENT v5

Confirmed role ownership and delivered paths

Role	Registered team member	Delivered on main
Role 1	Soha Raees	src/voice: AssemblyAI-compatible conversation, turn sanitisation, demo handoff and voice UI.
Role 2	Fazwan Zainuddin	src/extraction: schema-first extraction, safe failures, demo adapter and OpenAI-compatible provider.
Role 3	Mozzam Shahid	src/backend: FastAPI authentication, validation, SSE, tools, staff review and persistence handoff.
Role 4	Ishaan Sama	src/data + migration: synthetic data, Supabase/SQLite persistence, RBAC, encryption and audit.
Role 5	Jonathan	app/dashboard + components: live case list, gaps/conflicts, edits, review and approval workflow.
Demo	Mariam Habib	Issue #11 remains open: demo script, rehearsal, fallback assets and presentation delivery.

Ishaan Sama tech-stack input - implementation outcome

Contribution	Current implementation	Owner / readiness
AssemblyAI voice layer	Role 1 includes the AssemblyAI-compatible agent, browser audio/captions, turn-taking configuration and secure token route.	Role 1 + Role 3; live use needs a fresh AssemblyAI key.
Demo call transport	Browser voice UI is the MVP transport. Twilio is not required for the current competition path.	Role 1 + Role 5; local demo path verified.
Schema-first extraction	StructuredCase remains schema-first with sources, gaps, conflicts and safe failure. OpenRouter/OpenAI is optional; demo fallback exists.	Role 2 + Role 3; tests pass.
Distress indicator	Safety signals remain separate from final clinical triage. No autonomous diagnosis or final decision is permitted.	Roles 2/3/5; domain vocabulary still needs #6 sign-off.
Mock patient data	Supabase/Postgres is the hosted store; SQLite is the local/test fallback. The Role 4 migration creates case/review/audit tables.	Role 4 + Role 3; confirm fresh-key access and migration.
Brief and dashboard	FastAPI publishes SSE updates to the Next.js staff dashboard; review actions return through Role 3 and save through Role 4.	Roles 2/3/4/5; automated and local smoke tests pass.
Privacy and scope	Synthetic data, RBAC, encryption, retention and audit controls are implemented. Secrets stay in an untracked local .env.	All roles; exposed keys must be rotated before live demo.

REMAINING READINESS GATES

Rotate exposed keys; configure a clean local .env; resolve public-deployment dependency risk; verify Supabase and one approved save; close #6, #7 and #11; rehearse.

Contribution source: Ishaan Sama, EmergencyVoice - Tech Stack Document (31 Aug 2026). Current evidence: main aa95e0c, local verification and PR #34 test report (8 Sep 2026).